CS2840 Lecture 13: OT Extensions

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1 Introduction

This lecture surveys several important *extensions* of Optimal Transport (OT) obtained by relaxing the standard assumptions (two normalized measures on the same space with a fixed coupling structure). We cover: multi-marginal OT, unbalanced OT, and the Gromov–Wasserstein distance.

Housekeeping. Project proposal due Oct 21; PSet 3 due Oct 24. See project guidelines for expected formats and select a theme.

Recap (from last time). OT as an ML loss; differentiability of OT (w.r.t. marginals, cost, couplings); automatic differentiation pipelines (PyTorch/JAX); minibatching; sliced OT (leveraging the closed-form 1D OT solution).

Motivation. Starting from OT between two normalized measures on the same space, we relax: (i) the number of measures (two → many), (ii) mass conservation (equal total mass → unequal mass), and (iii) ambient space (same → different spaces).

Limitation	Extension	Example
2 measures only	Multi-marginal OT	Barycenters
Equal total mass	Unbalanced OT	Single-cell data
Fixed coupling structure	OT + constraints	Finance (martingale)
Same ambient space	Gromov–Wasserstein	Shape/graph matching
Scalar measures	Vector/Matrix OT	Tensors/Covariances

Table 1: Different flavors of OT. Many principles are shared across these flavors.

See the atlas in Fig. 1 for a broader landscape.

2 Multi-Marginal OT (MMOT)

2.1 Motivation

Many problems involve $S > 2$ measures: computing barycenters; multi-agent interactions (each agent has a distribution and we seek a consensus/equilibrium); physics/chemistry (interacting molecules/particles). The shared principle is to find the best "agreement" among measures.

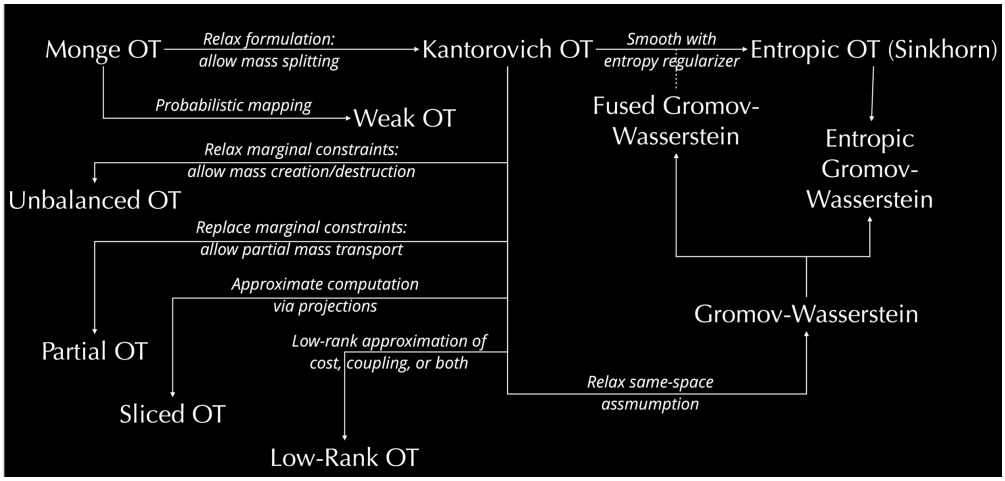


Figure 1: OT Atlas (refer to textbook for items not covered here).

2.2 Formulation

For general measures $\rho_1, \dots, \rho_S$ on spaces $X_1, \dots, X_S$ with a cost function $c : X_1 \times \cdots \times X_S \to \mathbb{R}$,

$$\min_{\pi \in \mathcal{M}(X_1 \times \cdots \times X_S)} \int_{X_1 \times \cdots \times X_S} c(x_1, \dots, x_S)\, d\pi(x_1, \dots, x_S) \qquad \text{s.t. } (\mathrm{Proj}_{X_s})_{\#}\pi = \rho_s,\ \forall s. \tag{1}$$

In the discrete case, with histograms $a^{(s)} \in \mathbb{R}_+^{n_s}$ and tensor cost $C \in \mathbb{R}^{n_1 \times \cdots \times n_S}$,

$$\min_{\pi \geq 0} \langle C, \pi \rangle \quad \text{s.t.} \quad \sum_{i_1, \dots, i_S :\, i_s = \ell} \pi_{i_1, \dots, i_S} = a_\ell^{(s)} \quad \forall s, \ell. \tag{2}$$

This recovers classic Kantorovich OT when $S = 2$.

Feasible set. For continuous measures, $\Pi\big((\rho_s)_{s=1}^S\big) = \{\pi \in \mathcal{M}(X_1 \times \cdots \times X_S) : (\mathrm{Proj}_{X_s})_{\#}\pi = \rho_s\}$. For discrete measures, $\Pi\big((a^{(s)})_{s=1}^S\big) = \{\pi \in \mathbb{R}_+^{n_1 \times \cdots \times n_S} :\ \sum_{i_1, \dots, i_S :\, i_s = \ell} \pi_{i_1, \dots, i_S} = a_\ell^{(s)}\}$.

2.3 Barycenters as MMOT

The Wasserstein barycenter of $\{\mu_s\}_{s=1}^S$ with weights $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_S)$ solves

$$\nu^\star \in \arg\min_{\nu} \ \sum_{s=1}^S \lambda_s\, W_p^p(\mu_s, \nu). \tag{3}$$

By introducing ν as an $(S+1)$-st marginal and applying the gluing lemma, one can reduce to a single $(S+1)$-way problem

$$\min_{\pi} \ \int_{X_1 \times \cdots \times X_S \times X} \Big(\sum_{s=1}^S \lambda_s\, c(x_s, x) \Big)\, d\pi(x_1, \dots, x_S, x) \quad \text{s.t. } (\mathrm{Proj}_{X_s})_{\#}\pi = \mu_s,\ \forall s, \tag{4}$$

and after minimizing w.r.t. ν one obtains the S-way cost

$$c(x_1, \dots, x_S) = \sum_{s=1}^S \lambda_s\, c\big(x_s, \mathrm{B}_{\boldsymbol{\lambda}}(x_1, \dots, x_S)\big), \qquad \nu^\star = (\mathrm{B}_{\boldsymbol{\lambda}})_{\#}\pi. \tag{5}$$

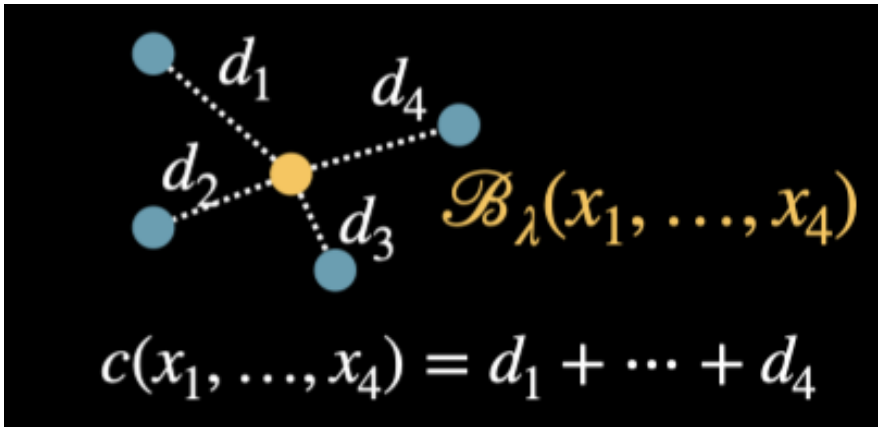


Figure 2: Barycenter intuition: the coupling acts as a "meeting point" among sources.

2.4 Intuition and Computation

The value $\pi(x_1, \ldots, x_S)$ represents *consensus mass* for the tuple across sources; if each measure places high mass near that tuple, the consensus is high.

With entropic regularization $\varepsilon > 0$,

$$\min_{\pi} \langle C, \pi \rangle - \varepsilon \, \mathsf{H}(\pi), \qquad \pi_{i_1,\ldots,i_S} = K_{i_1,\ldots,i_S} \prod_{s=1}^{S} u_{i_s}^{(s)}, \tag{6}$$

and generalized Sinkhorn updates for $s = 1, \ldots, S$ read

$$u^{(s)} \leftarrow a^{(s)} \oslash \Big(K \times_{r \neq s} u^{(r)} \Big), \tag{7}$$

where $\times_{r \neq s}$ denotes tensor contraction over all modes except s and $\oslash$ is elementwise division.

Question

Dynamic OT: can we define a vector field that converges to the barycentric center? (More subtle, but can be formalized.)

2.5 Applications

- **Multi-marginal Wasserstein GAN (Cao et al.)** — Learns separate distributions for facial attributes (pose, lighting, expression) and combines them via a barycentric consensus to synthesize coherent faces.
- **Deep Multi-Marginal Momentum Schrödinger Bridge (Chen et al.)** — Matches multiple marginals along a stochastic process using MMOT couplings with momentum regularization for stability.
- **Quantum chemistry (DFT)** — Captures multi-electron interactions via multi-way couplings that encode correlated electron densities beyond pairwise transport.
- **Fluid mechanics** — Fuses multi-sensor or temporal distributions to infer consensus flow fields consistent with transport dynamics.

3 Unbalanced OT (UBOT)

3.1 Motivation

In biology (e.g., single-cell trajectories under treatment), measurements are destructive; populations across time are different but *distributionally comparable*. Some cells die or proliferate under treatment, so total mass differs between source and target, violating classic OT's mass constraint. A softer, penalized relaxation is needed.

3.2 Formulation

Given histograms $a \in \mathbb{R}^n_+$ and $b \in \mathbb{R}^m_+$ that need not sum to the same total, relax marginal constraints with divergences D_ψ:

$$\min_{\pi \geq 0} \langle C, \pi \rangle + \tau_a \, D_\psi(\pi \mathbf{1}_m \,\|\, a) + \tau_b \, D_\psi(\pi^\top \mathbf{1}_n \,\|\, b). \tag{8}$$

3.3 Limit behavior (useful heuristics)

- $\tau \to \infty$: $\xi = \frac{\tau}{\tau+\varepsilon} \to 1$, recovering balanced OT with exact marginals.
- $\tau \to 0$: $\xi \to 0$, preferring divergence penalties over transport; approaches divergence-based measures (e.g., Hellinger).
- **Smaller** τ: leads to larger deviations from marginals a, b, good for handling noisy or partial data.
- $\varepsilon \to 0$ **caveat**: formally enforces marginals, but multiplicative form $\pi = \mathrm{diag}(u) K \mathrm{diag}(v)$ fails; handle with care.

3.4 Intuition and Unbalanced Barycenters

UBOT is necessary when $\sum_i a_i \neq \sum_j b_j$, but even when totals match, UBOT helps regularize outliers by tolerating marginal violations if they are cheaper than transportation. For barycenters, UBOT allows smooth splitting/merging of mass without brittle bifurcations.

Question

Can we combine unbalanced OT with multi-marginal OT? **Yes**—the relaxation is orthogonal and composes with other flavors.

3.5 Computation

With entropy $\varepsilon > 0$ and kernel $K = e^{-C/\varepsilon}$, modified Sinkhorn iterations read (elementwise operations):

$$u \leftarrow (a \oslash (Kv))^{\xi_a}, \quad \xi_a = \tfrac{\tau_a}{\tau_a+\varepsilon}, \tag{9}$$

$$v \leftarrow (b \oslash (K^\top u))^{\xi_b}, \quad \xi_b = \tfrac{\tau_b}{\tau_b+\varepsilon}, \tag{10}$$

yielding $\pi = \mathrm{diag}(u)\, K\, \mathrm{diag}(v)$ *without* enforcing exact marginals. Exponents yield partial rescaling, trading off marginal fidelity vs. transport.

Question

Notes: (i) View UBOT as clustering small OT problems and discouraging cross-cluster movement; (ii) sometimes cheaper to pay marginal penalties than shipping costs.

4 Gromov–Wasserstein (GW)

4.1 Motivation

To compare distributions on *different* spaces (different geometry, dimensionality, or global orientation), GW compares *intrinsic* structures. Examples: aligning neural activations (different seeds lead to rotated subspaces); unsupervised word translation using co-occurrence geometry. Plain Euclidean alignment fails up to rotations.

Question

Distances between embeddings in different languages are meaningless up to rotations; joint training can align spaces to expose consistent structures.

4.2 Formulation

Let metric-measure spaces (X, d_X, α) and (Y, d_Y, β) with pairwise distances $d_X(x, x')$ and $d_Y(y, y')$. The (squared) GW problem with loss $\ell(t, t') = (t - t')^2$ is

$$GW^2\big((\alpha, d_X), (\beta, d_Y)\big) \;:=\; \min_{\pi \in \Pi(\alpha,\beta)} \iint \big(d_X(x, x') - d_Y(y, y')\big)^2 \, d\pi(x, y) \, d\pi(x', y'). \tag{11}$$

In discrete form, with distance matrices $D \in \mathbb{R}^{n \times n}$ and $D' \in \mathbb{R}^{m \times m}$,

$$\min_{\pi \in \Pi(a,b)} \sum_{i,i',j,j'} \big|D_{ii'} - D'_{jj'}\big|^2 \, \pi_{ij} \pi_{i'j'} \;=:\; \min_{\pi} \; \mathcal{E}_{D,D'}(\pi). \tag{12}$$

GW defines a metric on isomorphism classes of metric-measure spaces (zero iff spaces are isometric), is symmetric, satisfies the triangle inequality, and is invariant to rigid transforms.

4.3 Computation

The objective is non-convex (quadratic in π) and relates to the NP-hard quadratic assignment problem. Entropic regularization convexifies a local surrogate and enables Sinkhorn-like iterations (Solomon et al., 2016): at iteration t,
1. Form a local linearization $C^{(t)} \leftarrow -D \, \pi^{(t)} \, D'$;
2. Update $\pi^{(t+1)} \leftarrow \arg\min_{\pi \in \Pi(a,b)} \langle C^{(t)}, \pi \rangle - \varepsilon \, \mathsf{H}(\pi)$ via Sinkhorn.
This can be interpreted as mirror descent; beware memory costs of storing (D, D', π).

Question

GW is useful for model distillation and OT model fusion when weights are unavailable.

4.4 Applications

- **Unsupervised word translation (AM & Jaakkola, 2018)** — Aligns languages using intra-language geometry derived from co-occurrence embeddings; similar words remain close post-alignment.
- **Universal geometry signals** — Recurrent semantic structures (e.g., 7 days, 12 months) act as geometric anchors for alignment.
- **Predictive of downstream performance** — GW distance correlates with translation accuracy and performs competitively at a fraction of adversarial training cost.
- **Semantic change over time** — Align embeddings across time slices to observe meaning drift without explicit dictionaries.
- **Linguistic distance** Estimate similarity between languages (e.g., Russian similar to Spanish, far from Chinese) through GW alignment of embedding geometries.

5 Takeaways

- Many OT extensions arise by relaxing classic assumptions.
- Entropic regularization is a powerful computational tool.
- Similar algorithmic ideas recur across flavors.
- Next time: guest lecture by Katy Craig on Dynamic OT.